Group Project: Mathematical Modelling

Analysis, Simulation and Prediction

Project at a Glance

Group size: 5 students **Total groups:** 18 **Marks:** 10 marks,
Each group must investigate a **distinct real-world problem via mathematical model**. Proposed topics must be approved to avoid substantial duplication. Projects may be based on a mathematical model arising from a biological system, such as infectious disease transmission, population dynamics, predator–prey interactions, species competition, ecological systems, etc.

1. Marking Scheme

Assessment criterion	Marks
LaTeX report: Submit an editable `.tex` file, compiled PDF, references, and figure files in `.eps` format. Include an introduction, motivation, objectives, assumptions, mathematical formulation, relevant diagrams, mathematical analysis (e.g., equilibria, stability, sensitivity, or other appropriate properties), numerical simulations, graphs, and interpretation.	**3**
Jupyter Notebook: Submit an executable `.ipynb` file with detailed, colourful Markdown explanations, mathematical equations, and algorithms *before* their respective code sections, along with labelled plots and interpretations.	**2**
Presentation and individual understanding: Clearly present the motivation, model, mathematical reasoning, implementation, results, and limitations. Every member must explain their contribution and answer questions. Usual presentation must be completed in 10 min and 5 min for question-answers	**5**
Total	**10**

2. Important Instructions

1. All 18 groups must work on distinct topics or modelling approaches, subject to prior approval.
2. Published models may be used as a starting point, but each group must demonstrate its own mathematical analysis, implementation, interpretation, or meaningful extension.
3. Select analytical techniques appropriate to the model. For example, equilibrium and stability analysis are suitable for many dynamical systems, while data-driven models may require different methods.
4. AI tools may be used; however, students must be able to **explain every equation, numerical method, algorithm, and code component** submitted.
5. Acknowledge all AI platforms used (e.g., ChatGPT, Gemini, Claude, etc.), specifying their roles. Cite all external datasets, published results, figures, and adapted material.
6. Any dishonesty in acknowledging the resources or AI tools used may cause you a penalty of 10 marks.
7. **Conceptual understanding, mathematical reasoning, and meaningful interpretation** are more important than cosmetic complexity.

3. Optional Advanced Component - Up to 5 Bonus Marks

Possible approaches include advanced numerical methods, machine learning, physics-informed neural networks (PINNs), data-driven forecasting, etc.
Groups may use **publicly available real-world datasets** relevant to their selected problem. To be considered for bonus marks, they should, *where applicable*:

- Cite verifiable data sources and clearly explain the advanced modelling or computational methodology.
- Validate results against independent observations, benchmark solutions, published findings, or other appropriate evidence; for prediction tasks, use data not included in training or calibration.
- Report suitable quantitative measures, such as MAE, RMSE, relative error, convergence rates, or optimality gaps, and provide appropriate comparison graphs.
- Explain the underlying mathematics, algorithms, assumptions, implementation, and limitations thoroughly.

Note: Merely using AI tools, sophisticated software, or presenting predictions does not automatically qualify a group for bonus marks. Bonus marks depend on mathematical depth, independent effort, and credible validation.

4. Final Instructions

1. **Presentation Schedule:** Reports may be presented during regular lecture/tutorial hour or permitted slot outside the lecture hours.
2. **Prior Approval:** To avoid clashes, the presentation topic, date, and time must be finalized in consultation with me on or before October 19, 2026. Slots date, time and topics will be allotted on a first-come, first-served basis and in the order filled in the sheet. Please fill the Google sheet for it:

   https://docs.google.com/spreadsheets/d/1mdnSD2K1HQwVpAh7lvSyaqCFLnAjwh54dGs0xQHoIUs/edit?usp
3. **Final Deadline:** All project presentations must be completed **before 14th November 2026**. The available time slots are given in subsheet in the above google sheet.
4. **Report Submission:** The project .tex file with all figures, its pdf and Jupyter notebook .ipynb file must be submitted to me **at least one week before the scheduled presentation**.